

PD Model Validation Report

1. Introduction and Analysis Approach

This report presents a basic validation of the provided Probability of Default (PD) model outputs using historical data for the years 2022–2024.

The goal of the analysis was to understand and validate the given model outputs. The main focus was on:

- checking data structure and consistency,
- assessing data quality and missing values,
- verifying PD and PD pool logic,
- evaluating whether PD estimates reflect observed default risk,
- reviewing model behaviour over time.

The analysis was performed using SQL for data preparation and quality checks and Python (pandas, matplotlib) for analysis and visualization.

2. Data Overview and Preparation

2.1 Data Sources

The analysis used the following datasets:

- dat_scoring.csv – model outputs at contract level (PD, PD pool, default flag),
- dar_arrangement.csv – contract-level financial and behavioural attributes,
- dat_obligor.csv – customer-level demographic information,
- par_pools.csv – PD interval thresholds for pool assignment,
- par_model.csv – risk attribute and coefficient structure of PD model used.

2.2 Initial Data Checks

Before starting validation, basic structural checks were performed:

- column types were reviewed and aligned to ensure correct joins and calculations,
- key identifiers (YEAR, AR_ID, IP_ID) were verified,
- duplicate records were checked at contract-year and customer-year level.

No duplicate records were found in the scoring data. Some duplicates were present in the arrangement data, which is consistent with multiple snapshots within a year. For analysis, a single representative record per year with the highest number of non-missing values was selected.

After cleaning and joining, a consolidated analysis dataset was created and used in all further steps.

3. Portfolio Overview and Key Observations

The portfolio covers three years (2022, 2023, 2024) with a same number of contracts each year. Portfolio size is therefore stable over time.

Key observations:

- The observed default rate is similar in 2022 and 2023, but increases significantly in 2024.
- Average PD also increases over time, showing consistent direction with observed defaults.

These observations guided the deeper analysis of PD pools and missing values.

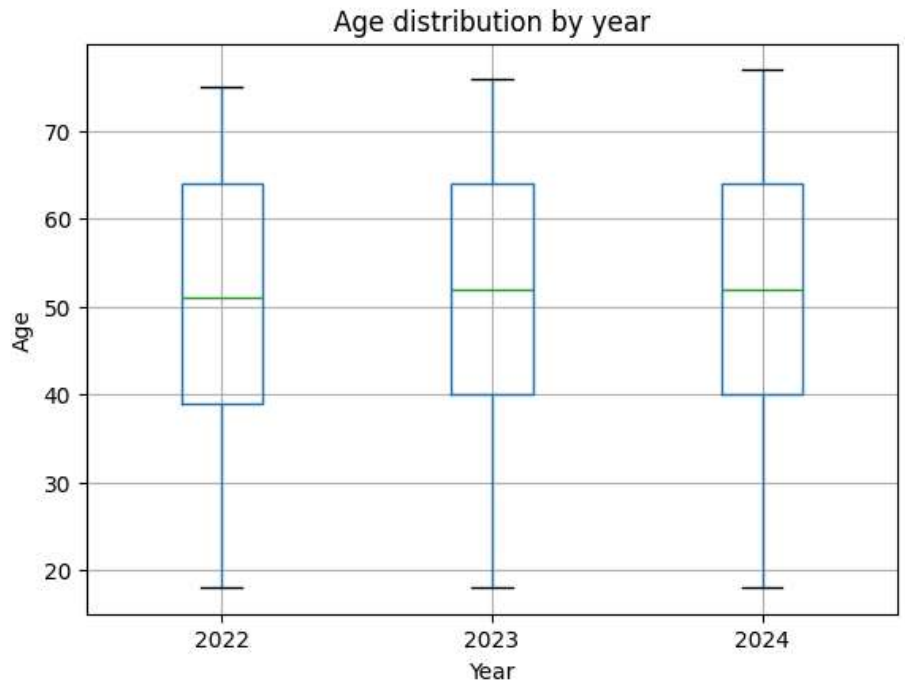
4. Portfolio Visualizations

Visual analysis was used to support qualitative assessment of portfolio structure, composition, and performance.

4.1 Obligor Characteristics

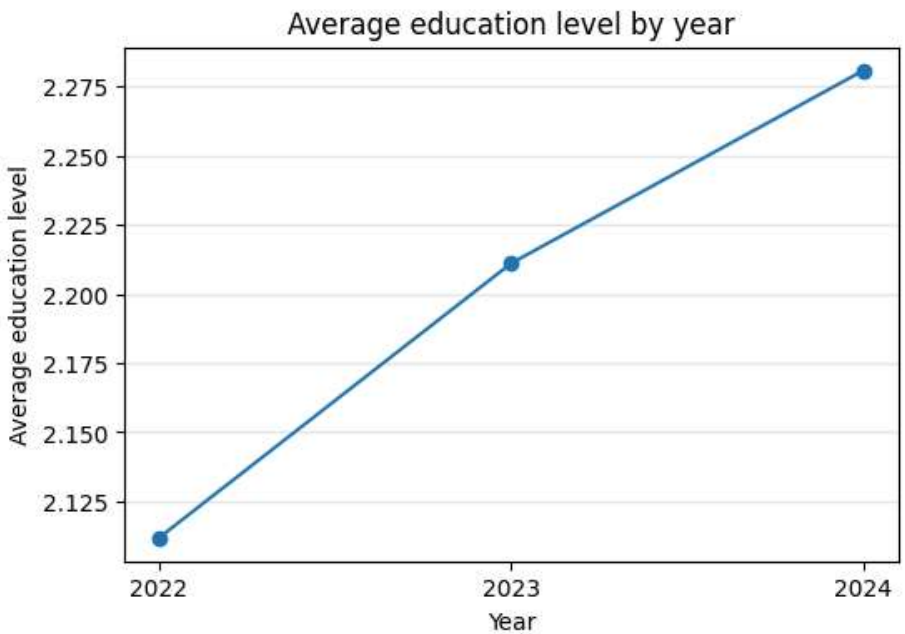
Age distribution by year (boxplot)

Shows stable age distribution across years, with no major structural shifts.



Average education level by year (line chart)

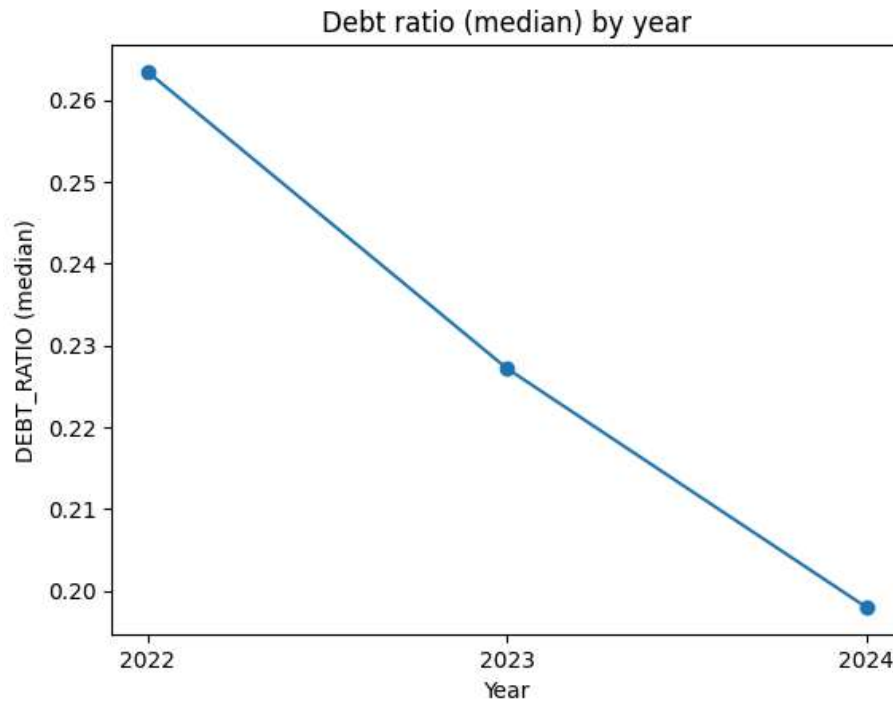
Displays a gradual increase in average education level, suggesting minor changes in customer composition over time.



4.2 Arrangement and Portfolio Features

Debt ratio (median) by year

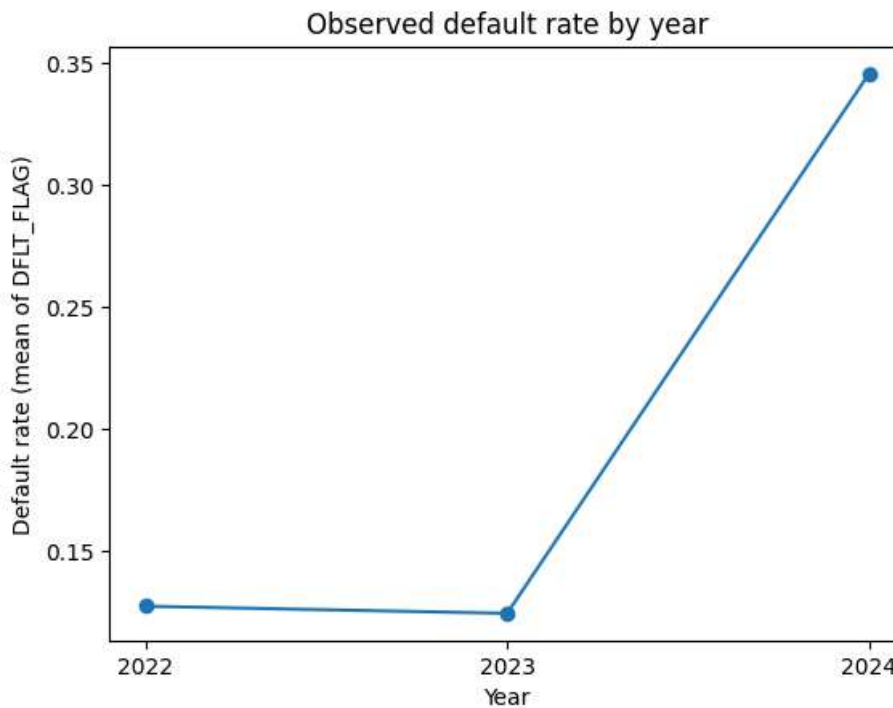
The median debt ratio shows moderate changes across years, with no extreme shifts. This suggests stable leverage at portfolio level.



4.3 Portfolio Performance

Observed default rate by year

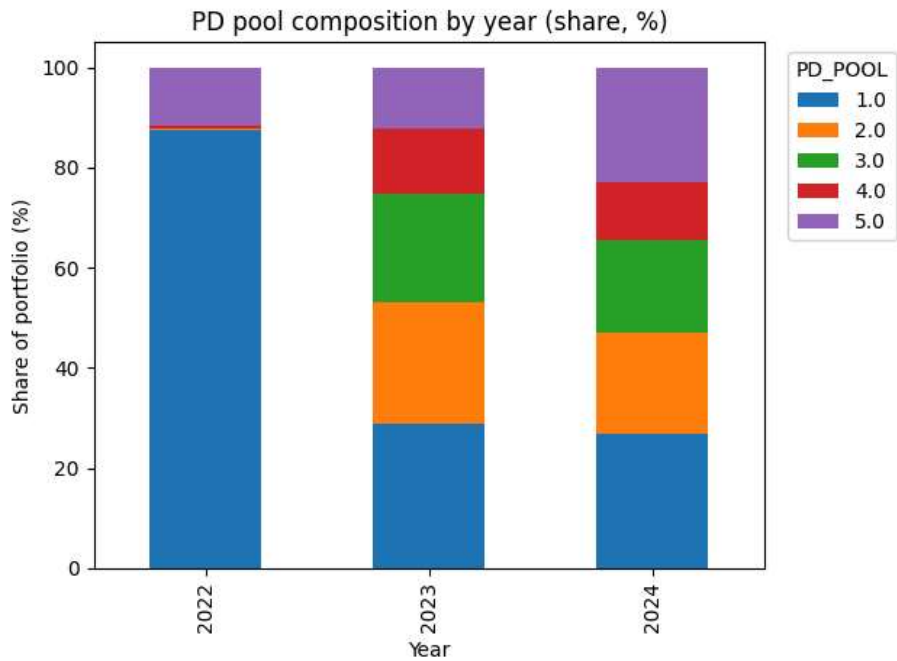
Default rates are stable in 2022–2023 and increase sharply in 2024, indicating higher overall portfolio risk in the latest year.



4.4 PD Pool Structure and Performance

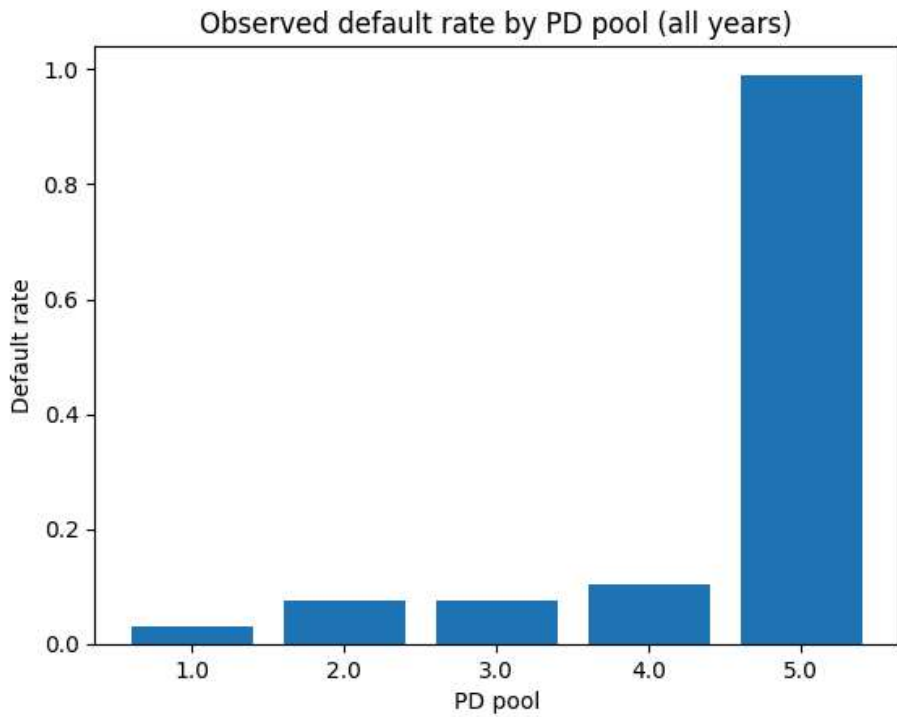
PD pool composition by year (share, %)

Shows how contracts are distributed across PD pools over time. The portfolio composition changes gradually, with higher-risk pools gaining share in 2024.



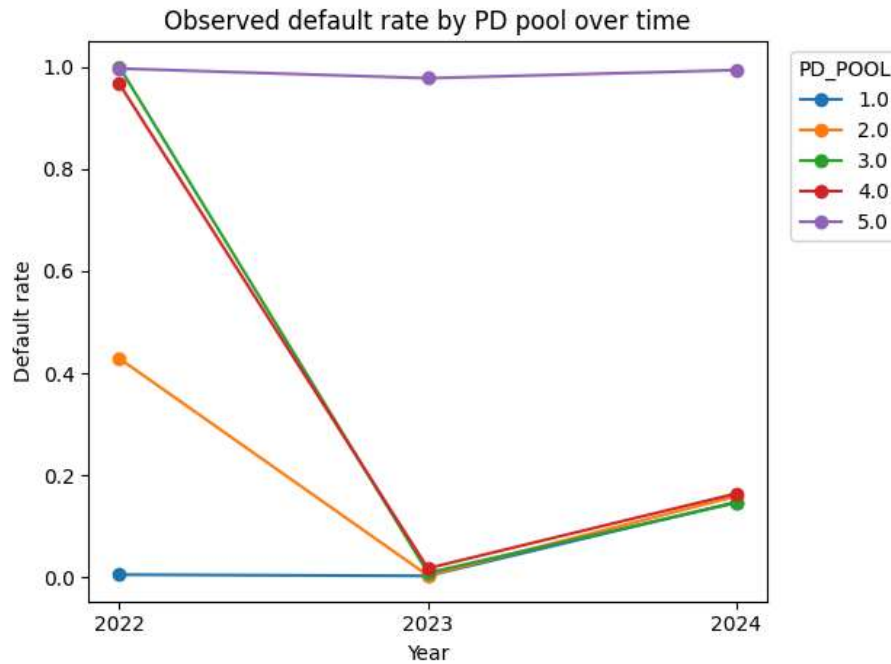
Observed default rate by PD pool (all years)

Higher PD pools consistently show higher default rates, confirming that PD pools separate risk levels.



Observed default rate by PD pool over time

Pool 5 remains a stable high-risk group, while lower pools show stronger changes over time.



5. Data Quality Assessment

5.1 Completeness

Key identifiers (YEAR, AR_ID, IP_ID) are fully populated, ensuring reliable joins and aggregation.

Arrangement-level behavioural variables (DPD, M_LAST_DPD) have many missing values, which is expected as they are only present when payment delays occur.

PD values are missing for a small share of the portfolio (~2–3%).

Obligor-level demographic data is missing for a larger share of observations (~14%), which is common in real-world data.

5.2 Missing Value

Missing values were analysed to understand whether they introduce bias:

Records with missing DPD values show significantly lower observed default rates than records with non-missing DPD. This indicates that DPD missingness is informative and aligned with business logic, rather than random.

Records with missing PD values exhibit default rates similar to those with available PD values. This suggests that missing PD values do not introduce material bias into portfolio-level conclusions.

Review of the model parameter file confirms that missing values are explicitly included in the model structure and treated as a separate category. This supports the observed missing patterns in behavioural variables and indicates that missing values are part of the intended model design rather than data quality issues.

Overall, missingness is not random, but does not invalidate model conclusions.

5.3 Consistency

Missing values are evenly distributed across years.

No year-specific data breaks or anomalies were detected.

Data behaviour is consistent with expected business logic.

6. Assessment of PD Calculation Correctness

PD calculation correctness was assessed by verifying PD pool assignment logic.

For each available PD value, the expected PD pool was recalculated based on the provided PD interval thresholds and compared with the original PD pool.

Results:

- All records with available PD values were assigned to the correct PD pool.
- All observed mismatches were explained by missing PD values.
- No systematic misclassification was identified.

This confirms that PD pool assignment is correct and consistent with model rules.

7. PD Model Performance and Stability Over Time

7.1 Risk Differentiation

The model clearly differentiates risk:

- Lower PD pools have lower observed default rates.
- Higher PD pools have higher observed default rates.
- Pool 5 consistently represents the highest-risk segment.

This confirms model clear risk separation.

7.2 Stability Over Time

PD pool risk ordering remains logical across years. However, observed default rates increase across most pools in 2024. This suggests a general risk increase, potentially due to macroeconomic effects or reduced model calibration over time. Given the available information, this is interpreted as portfolio risk change, not a calculation error.

8. Final Conclusions

- The PD model outputs are internally consistent and correctly calculated.
- Data quality issues are present but expected and manageable.
- Missing values do not introduce critical bias.
- PD pools clearly separate risk and reflect observed default behaviour.
- An increase in default rates is observed in 2024 across the portfolio.

This analysis supports the validity of the provided PD model outputs at a basic validation level.